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Roll No

BE-3001(CE)-CBGS

B.E., IV Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Mathematics - III

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) Find the Fourier series to represent the function
$$f(x) = |\sin x|, -\pi < x < \pi.$$

b) Expand $f(x) = \sqrt{1 - \cos x}$, $0 < x < 2\pi$ in a Fourier Series.
2. a) Find Fourier sine transform of the function
$$2e^{-5x} + 5e^{-2x}.$$

b) Find the Fourier transform of $f(x) = \begin{cases} 1, & |x| < a \\ 0, & |x| > a \end{cases}$
3. a) Find $L\{F(t)\}$, if $F(t) = \begin{cases} (t-1)^2, & t > 1 \\ 0, & 0 < t < 1 \end{cases}$
b) By convolution theorem evaluate $L^{-1}\left\{\frac{1}{(s^2 + 9)(s + 3)}\right\}.$

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4. a) Use convolution theorem to find

$$L^{-1}\left\{\frac{1}{(p+1)(p-2)}\right\}$$

- b) Find Laplace transform of the followings:

i) $L\{e^t \sin^2 t\}$ ii) $L\{t^2 \sin at\}$

5. a) Use Cauchy-Riemann equation to find v when

$$u = 3x^2y - y^3.$$

- b) Evaluate the integral $\int_0^{1+i} z^2 dz$

6. a) Use Cauchy's integral formula, evaluate the integral

$$\int_C \frac{e^{2z}}{(z-1)(z-2)} dz, \text{ where } C \text{ is the circle } |z| = 3.$$

- b) Show that $\int_0^{2\pi} \frac{d\theta}{5+3\sin\theta} = \frac{\pi}{2}.$

7. a) Using bisection method; find a real root of the equation $x^4 - x - 10 = 0.$

- b) Apply Newton's method to evaluate $\sqrt{12}$ approximately.

8. a) Apply Newton-Raphson method to solve $3x = \cos x + 1.$

- b) Find the quadratic factor $x^3 - 3.7x^2 + 6.25x - 4.069$, after two iteration, given $p_0 = -2.5$, $q_0 = 0$, by using Bairstow's method.

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